AI Based Cardiovascular Risk Assessment System Using Explainable Machine Learning

Research proposal

Study focus Development and evaluation of a web prototype for 10 year cardiovascular mortality

Prepared by Axror Nosirov, Jahongir Islomov and Abbosxoja Obidoxojayev

Date 29 October 2026

Status Proposal for completing and evaluating an existing research prototype

Abstract

We propose to complete and evaluate a reproducible web application that estimates 10-year cardiovascular mortality from baseline clinical and lifestyle variables. The study uses public US National Health and Nutrition Examination Survey data from 1999–2008 linked to the CDC 2019 public-use mortality files. The constructed cohort contains 11,843 adults aged 40–79 without self-reported prior cardiovascular disease. We use 8,902 participants for development and 2,941 from a later survey cycle for temporal evaluation. The fixed inputs are age, the NHANES recorded male/female category, systolic blood pressure, BMI, total cholesterol, HDL cholesterol, current smoking and self-reported diabetes.

A working React and FastAPI prototype, logistic baseline, fixed comparison with random forest and histogram gradient boosting, and a development-only calibration experiment already exist. This proposal sets out the remaining sensitivity analyses, explanation research, user evaluation and reporting. We will assess discrimination and calibration, preserve training-fold separation, document missing-data handling and communicate uncertainty without inventing individual confidence scores. The expected output is a transparent research system with reproducible evidence. Its current intended use is demonstration with synthetic cases. Independent validation and clinical governance would be required for patient-care use.

1 Background and rationale

Cardiovascular prediction requires an explicit population, outcome and prediction horizon. In this project, “risk” means cardiovascular death within ten years after a baseline examination. The endpoint includes fatal heart disease and cerebrovascular disease; it does not provide a diagnosis of existing disease or a count of nonfatal heart attacks and strokes. Linking survey measurements to mortality follow-up makes this target feasible without access to private hospital records. [1,2]

Our rationale has two parts. First, clinical measurements and questionnaire responses require harmonized definitions and explicit treatment of missing values before they become model inputs. Second, a prediction system must evaluate whether its probabilities agree with observed outcomes as well as whether it ranks participants correctly. Calibration is therefore a central research measure alongside discrimination. [5]

Explainability provides an additional view of how a fitted model uses recorded inputs. SHAP offers a framework for attributing individual model predictions to features. It does not by itself establish causal effects, validate the model or justify treatment decisions. We will study explanations together with the input-quality information that users need to interpret them. [6]

2 Problem statement

The project addresses three connected problems. Clinical and lifestyle data can contain inconsistent coding, missing measurements and extreme observations. A model can distinguish participants with different outcomes while assigning unreliable probabilities. Finally, a probability alone does not show which recorded factors shaped an individual result or whether missing inputs weakened its interpretation. Our study will examine these problems through a shared data pipeline, controlled model comparisons and a working explanation interface.

3 Aim and research objectives

The general aim is to develop and evaluate an explainable cardiovascular mortality research system that accepts validated baseline inputs, produces a clearly defined probability and presents model factors with appropriate limitations.

Research question	Specific objective
RQ1 How can validation and preprocessing integrate clinical and lifestyle data into reliable model inputs?	RO1 Build a reusable pipeline and quantify missingness, invalid inputs, imputation and completeness.
RQ2 How well do the candidate models discriminate and calibrate 10-year cardiovascular mortality?	RO2 Compare fixed models and calibration methods using development validation and a transparently reported later-cycle evaluation.
RQ3 How can explanations show the factors shaping an estimate and communicate uncertainty clearly?	RO3 Implement individual explanations and input-quality displays, then assess their fidelity and user understanding.

We will compare the models empirically without assuming that complexity improves performance. The contribution is a reproducible applied system and evaluation.

4 Study scope and data

4 1 Data sources

Baseline data come from the five NHANES cycles spanning 1999–2008. Examination, laboratory, demographic and questionnaire files are joined by participant identifier SEQN. The source manifest records 42 NHANES XPT files, their official URLs and checksums. The CDC 2019 public-use linked mortality release supplies follow-up through 31 December 2019. We will retain this documented release for reproducibility rather than silently substituting a newer linkage. [1,2,10]

Some public mortality records contain privacy-protecting substitutions for follow-up time or cause of death. The CDC states that vital status itself was not changed. We will disclose how these limitations affect horizon-specific and cause-specific interpretation. [1]

4 2 Population and outcome

Eligible participants completed the baseline examination, were aged 40–79, were eligible for mortality linkage and had a determinable ten-year outcome. The primary-prevention cohort requires definite negative answers for prior congestive heart failure, coronary heart disease, angina, myocardial infarction and stroke. Uncertain prior-history responses are excluded and counted in the cohort flow.

The binary target is cardiovascular death within 120 months after examination, using CDC underlying-cause groups 001 for diseases of heart and 005 for cerebrovascular disease. These are public-file cause categories, not ICD-10 codes. A known non-cardiovascular death before ten years is a non-event for this binary endpoint. Records without sufficient information to establish the target are excluded rather than labeled as survivors. [2]

Role	Cycles	Participants	Target events	Event rate
Development	1999–2006	8,902	274	3.08%
Temporal evaluation	2007–2008	2,941	87	2.96%

4 3 Model inputs

Predictor	Unit or coding	Definition or handling
Age	Years	Recorded baseline age, restricted to 40–79.
NHANES recorded category	Male or female	The two source survey codes do not represent the full range of gender identity.
Systolic blood pressure	mmHg	Mean of available positive systolic examination readings.
Body mass index	kg/m²	Positive recorded BMI, with extreme values flagged for review.
Total cholesterol	mg/dL	Baseline laboratory measurement.
HDL cholesterol	mg/dL	Harmonized source variable with documented assay changes across cycles.
Current smoking	Yes or no	Derived from lifetime and current smoking questions; unknown answers remain missing.
Self-reported diabetes	Yes or no	Borderline, refused and unknown responses remain missing.

LDL cholesterol, physical activity, medication and alcohol use are outside the current core model. Adding a predictor would require a documented cross-cycle derivation and an amended analysis plan. The web demonstration uses invented profiles and does not collect patient identifiers.

5 Research methodology

5 1 Design and prevention of leakage

We use a retrospective prediction-model study with a temporal split. Five stratified development folds generate out-of-fold probabilities. Imputation, scaling, encoding and any model tuning must use only the relevant training fold. Follow-up and cause-of-death fields are outcome labels and never predictor inputs. Changes to the cohort, variables, tuning or thresholds will be written as amendments before the new analysis runs.

The 2007–2008 cycle has already been inspected for the fixed logistic baseline. Subsequent model and calibration comparisons used development data only. We will disclose this history and avoid describing that cycle as a wholly unseen final test. New independent data would be required for stronger confirmation, especially in a local clinical population.

5 2 Data quality and missingness

The pipeline checks schema, units, valid source codes and explicit missing values. Numeric variables use training-fold median imputation and standardization. Recorded categorical and binary variables use training-fold most-frequent imputation and one-hot encoding. Missing values do not become zero automatically. In development data, systolic BP is missing in 4.46% of records and each cholesterol measure in 5.30%. All eight predictors are observed in 7,843 development participants. [10]

Remaining work will compare complete-case and imputed analyses, report participant and event losses, and assess whether conclusions change on comparable subsets. We will review influential extreme values and perform prespecified sensitivity analyses. HDL assay differences require cycle-level sensitivity checks, rather than treating observed shifts as a biological trend. Clinical review of hard input limits and softer check-entry warnings remains necessary. [4]

5 3 Candidate models and calibration

The current baseline is logistic regression with fixed L2 regularization and no class weighting. The fixed comparators are a conservative random forest and scikit-learn histogram gradient boosting. The first comparison is complete. Any additional tuning will use a limited, documented search within development data; the number of target events constrains model complexity. A neural network is not required for the current study.

We have compared the original logistic probabilities with sigmoid and isotonic calibration using five outer folds and three inner calibration folds. The outer assessment fold contributes to neither base-model fitting nor calibrator fitting. We retained the original output under a written rule requiring a Brier improvement with supportive uncertainty and better calibration diagnostics. This decision is preliminary and specific to the current development sample. [10]

5 4 Performance evaluation

Domain	Planned report
Discrimination	AUROC and AUPRC, with participant counts, event counts and event prevalence.
Probability quality	Brier score, calibration plot, mean predicted versus observed risk, joint calibration intercept and slope.
Precision	Intervals with the resampling method and its limitations stated explicitly.
Robustness	Missing-data, extreme-value and cycle sensitivity analyses with denominators.
Subgroups	Exploratory age and recorded-category comparisons only where event counts support interpretation.
Threshold performance	Sensitivity, specificity and predictive values only after a threshold and intended action are justified.

AUROC measures ranking, not classification accuracy. AUPRC must be interpreted alongside the uncommon outcome prevalence. Brier score measures probability error and also reflects discrimination, so it cannot establish calibration alone. A joint calibration slope near one and intercept near zero are useful diagnostics but do not guarantee agreement at every risk level. [5]

Current intervals resample people while holding predictions fixed. They omit model-fitting uncertainty and the complex NHANES design. Before making population-level claims, we will specify design-aware analysis with the retained weights, strata and primary sampling units. Combining the earliest cycles requires the supplied 1999–2002 four-year weights under CDC guidance. Until that work is complete, primary results describe the constructed sample. [3]

5 5 Explanations and uncertainty

The current app displays exact logistic feature contributions relative to the average transformed development input. Their scale is log-odds, not percentage-point risk change. Planned SHAP

work will document the explanation method, background data, output scale and handling of correlated inputs. We will check numerical fidelity and explanation stability across selected synthetic profiles. [6]

Model-level performance intervals and missing-input warnings will remain separate. We will explore bootstrap refitting to characterize model-fit variability for example predictions, but any display must state what the distribution represents. It must not imply a validated interval for an individual person's true risk. If the method cannot be justified, the interface will show known limitations and data completeness without an invented confidence rating.

A small formative comprehension study may use synthetic cases with intended users after the institution determines the required review and consent process. The protocol will specify recruitment, tasks and measures before data collection. Candidate measures include correct identification of the predicted outcome and horizon, understanding of missing-input warnings, and avoidance of causal interpretations. We will not claim a completed usability study or clinical benefit from this proposed work.

6 System development

The architecture connects a React interface to a FastAPI service. The service validates the eight fields, applies the saved preprocessing pipeline, predicts with the selected development model and returns factor explanations plus input-quality flags. The interface shows the exact outcome and horizon, research status, source population and printable synthetic-case report. It does not assign a clinical action threshold.

The first local implementation is complete. It accepts two synthetic preset profiles, supports missing optional measurements, rejects impossible values such as an out-of-scope age or negative blood pressure, and flags values outside the development data's observed range. Exact logistic contributions reconcile with the model score. Seven current API checks passed when the prototype was built and rerun. These software checks establish specified behaviors, not clinical validity. [10]

Remaining engineering work will strengthen error messages, accessibility, mobile behavior, report formatting and reproducibility. Model versions, preprocessing settings and dependency versions will be recorded. Raw and processed person-level data, serialized artifacts and submitted input values will not be committed inadvertently or exposed through a public demo.

7 Preliminary feasibility and current results

The following measurements are existing preliminary results, not promised outcomes of future work. All three candidate scores below use the same development participants and folds. [10]

Development model	AUROC	AUPRC	Brier score
Logistic regression	0.777	0.124	0.02841
Random forest	0.776	0.119	0.02855
Histogram gradient boosting	0.765	0.118	0.02866

Logistic regression has the lowest Brier score in this fixed comparison, although paired bootstrap intervals for differences include zero. On the later cycle, the fixed logistic baseline has AUROC 0.778 with a preliminary 95% interval of 0.738–0.819, AUPRC 0.080, and Brier score

0.0281. The evaluation cohort contains 87 target events. These unweighted measurements do not demonstrate local clinical performance.

The development calibration experiment produced Brier scores of 0.028412 for the original logistic output, 0.028434 for sigmoid and 0.028566 for isotonic. Sigmoid brought the joint calibration intercept and slope closer to their reference values but slightly worsened Brier. Neither adjustment met the written switching rule, so the demo retains the original model output.

8 Expected outcomes and acceptance criteria

The project will deliver an auditable cohort pipeline and data dictionary, a completed comparison and sensitivity report, a model card, an explanation and uncertainty study, and an editable research report following TRIPOD+AI. The software deliverable will be a working local research application with a reproducible model artifact and a synthetic-case demonstration. [7]

Completion requires a reproducible cohort flow; documented separation of development and evaluation; traceable model settings; performance tables with counts and limitations; tested input validation; and explanations whose stated scale matches their calculation. The final report will state whether each objective was achieved and where evidence remains insufficient. No minimum accuracy percentage or clinical benefit will be invented to define success.

9 Completion schedule and team responsibilities

The following is an indicative eight-week completion schedule beginning after proposal review. It describes future work and does not imply that the existing prototype took this amount of time. The team will adjust the schedule to institutional deadlines and available statistical or clinical advice.

Week	Primary activity	Deliverable
1	Verify literature and record analysis amendments	Source matrix and finalized sensitivity plan
2	Missingness, extreme-value and HDL sensitivity	Data-quality experiment report
3	Exploratory subgroup and survey-design work	Counts, precision review and documented limitations
4	SHAP fidelity and uncertainty exploration	Explanation module and method report
5	Interface, accessibility and report refinement	Reproducible application release
6	Approved formative comprehension study	User findings and revised wording
7	Reproducibility package and model card	Research tables and TRIPOD+AI draft
8	Final writing and demonstration rehearsal	Final report, presentation and demo

Proposed responsibilities are Axror Nosirov for data and modeling, Jahongir Islomov for backend integration and verification, and Abbosxoja Obidoxojayev for frontend, literature organization and report assembly. These are suggested allocations for team agreement. Every member should understand the outcome definition, evaluation design and limitations.

The current scope uses public data, open-source software and existing local computers. The tested model families do not require a GPU. Optional hosting, additional software services or

specialist advice may create costs; no external funding or price estimate is assumed. Access to private hospital records is not a prerequisite for completing the university prototype.

10 Ethics limitations and future clinical use

We will follow the public data's permitted use and the university's requirements for secondary-data research. Public availability does not establish the institution's ethics-review determination, so the team will obtain the applicable confirmation. Any comprehension study or future restricted-data work requires its own review and permissions. The research demo will use synthetic profiles and will not request names or patient identifiers. WHO guidance informs attention to human oversight, privacy and accountability. [8]

The principal limits are the US source population, historic baseline measurements, privacy substitutions in public mortality records, HDL method changes, incomplete predictors, self-reported history and the small number of later-cycle events. The current sample metrics are unweighted. A temporal split within NHANES does not establish transportability to Uzbekistan or another clinical setting. Explanations describe fitted associations, and the present study cannot estimate treatment effects.

The final university system will remain a research demonstration unless a separate clinical translation study supports a defined intended use. That pathway would require independent validation in the intended population with suitable outcomes, clinical review of inputs and decisions, privacy governance, assessment of applicable regulatory requirements, and an authorized prospective evaluation with monitoring. Manually entering real-patient values still constitutes use of real health data. IMDRF clinical-evaluation guidance provides a framework for assessing clinical claims, but this proposal does not claim device approval or authorize patient-care deployment. [9]

References

[1] Centers for Disease Control and Prevention. National Death Index and linked mortality files. Public-use 2019 linkage information. Accessed 29 October 2026. Source

[2] National Center for Health Statistics. 2019 public-use linked mortality files data dictionary. 2022. Source

[3] Centers for Disease Control and Prevention. NHANES tutorials on weighting and combining survey cycles. Accessed 29 October 2026. Source

[4] Centers for Disease Control and Prevention. NHANES 2007–2008 HDL cholesterol documentation and analytic notes on assay methods. Source

[5] Van Calster B, McLernon DJ, van Smeden M, et al. Calibration: the Achilles heel of predictive analytics. BMC Medicine. 2019;17:230. doi 10.1186/s12916-019-1466-7. Source

[6] Lundberg SM, Lee SI. A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems. 2017;30. Source

[7] Collins GS, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378. doi 10.1136/bmj-2023-078378. Source

[8] World Health Organization. Ethics and governance of artificial intelligence for health: WHO guidance. Geneva. 2021. ISBN 9789240029200. Source

[9] International Medical Device Regulators Forum. Software as a Medical Device (SaMD): Clinical Evaluation. IMDRF SaMD WG N41. 2017. Source

[10] Project team. Data dictionary, source manifest, cohort feasibility report, baseline results, development model comparison, calibration report and prototype source code. Internal reproducible research outputs. October 2026. These project results are preliminary and have not undergone independent peer review.